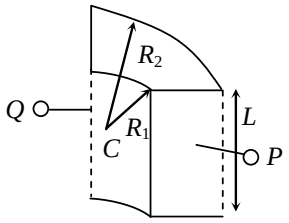
PHYSICS

SECTION – I
Single Correct Choice Type

This section contains 6 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

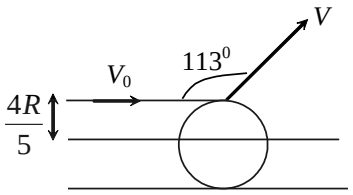
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1. A resistor is formed in the shape of a hollow, quarter cylinder from a material of resistivity ρ. The length of cylinder is L, inner and outer radii are R_1 and R_2 respectively. The resistance of this resistor between the shown terminals P and Q is



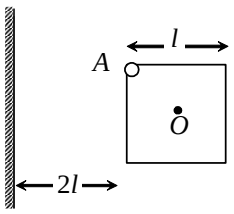
(a) $\dfrac{2\pi\rho L}{\left(R_2^2 - R_1^2\right)}$ (b) $\dfrac{2\rho\pi}{L\ln\left(\dfrac{R_2}{R_1}\right)}$ (c) $\dfrac{\pi\rho}{2L\ln\left(\dfrac{R_2}{R_1}\right)}$ (d) $\dfrac{\rho\pi R_1}{L(R_2 - R_1)}$

2. A smooth ball of steel is moving in horizontal direction with the speed of $V_0 = 5\sqrt{gR}$ where R is the radius of a smooth sphere fixed on the horizontal surface as shown in figure. The ball collides with the sphere and moves with changed velocity V along the line shown in figure. Then which of the following statements is correct?



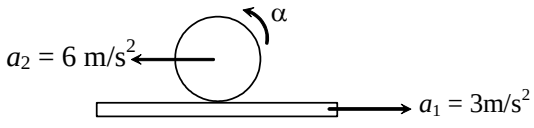
(a) $V = V_0$ (b) $4V = 5\,V_0$ (c) $8V_0 = 5\sqrt{3}V$ (d) $5V_0 = 8\sqrt{3}V$

3. A uniform wooden square board of side l is kept on a frictionless horizontal surface such that its distance from a vertical wall is $2l$ as shown in figure. A man having mass equal to the board is standing on the corner A of the board. If he moves to the centre O of the board then the distance of the centre of the board from the wall will become



(a) $\dfrac{\left(5\sqrt{2}-1\right)}{2\sqrt{2}}l$ (b) $\dfrac{\left(5\sqrt{2}\right)}{4}l$ (c) $\dfrac{5l}{2}$ (d) $\dfrac{9l}{4}$

4. In the following figure, a sphere of radius 3m is in pure rolling on the plank. The accelerations of the sphere and the plank are indicated. The value of angular acceleration α is

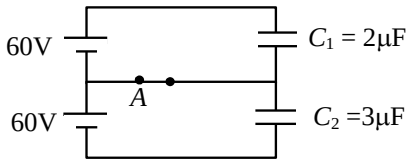


(a) 3 rad/s^2 (b) 6 rad/s^2

(c) 3 rad/s^2 (opposite to the direction shown in figure)

(d) 1 rad/s^2

5. Initially the switch *A* is open in the adjacent figure. Find the charge which flows through switch *A* after it is shorted.



(a) 24 μC (b) 36 μC

(c) – 36 μC (d) 60 μC

6. Standing waves are set up in a sting of length 240 cm clamped horizontally at both ends. The separation between any two consecutive points where displacement amplitude is $3\sqrt{2}$ cm is 20 cm. Standing waves were set by two traveling waves of equal amplitude of 3 cm. The overtone in which the string is vibrating will be

(a) 2nd (b) 3rd (c) 4th (d) 5th

SECTION – II
Multiple Correct Choice Type

This section contains 5 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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7. A long plank of mass M = 20 kg can move on horizontal frictionless surface freely. Initially the plank is at rest and an object of mass m = 5kg is given a velocity of 25 m/s and placed on the floor of the plank. Co-efficient of friction between object and floor of the plank is $\frac{5}{8}$. After having some relative movement w.r.t. the plank, the object finally starts moving with the plank in one piece. Then

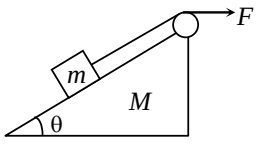
(a) the total work done by friction is –1250 J

(b) the minimum length of the plank should be 40 metres

(c) final velocity of plank is 10 m/s

(d) for time t = 3.2 seconds, the object is having relative motion w.r.t. plank

8. There is no friction anywhere in the system shown. The pulley is light and the wedge is free to move on the friction less surface. A horizontal force F is applied on the system in such a way that m does not slide on M or both move together with some common and constant acceleration. Given $M > m\sqrt{2}$



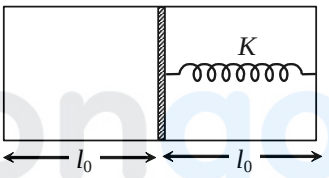
(a) Pseudo force acting on m as seen from frame of M is equal to $\frac{MF}{M+m}$ and is less than $mg \sin\theta$.

(b) Pseudo force acting on M as seen from the frame of m is greater than $\frac{mF}{m+M}$ and less than $mg \sin\theta$.

(c) Normal force between ground and M is greater than both $\frac{mF}{m+M}$ and mg sin θ

(d) Normal force (for θ = 45°) between m and M is equal to $\frac{mF}{m+M}$

9. A diathermic piston of mass M, cross section area A separate the volume inside a horizontal adiabatic cylinder of length $2l_0$ in two equal parts. Each chamber contains an ideal gas and pressure on each side is P. The piston can move without friction and is attached with a spring of spring constant K as shown. Initially the spring is non deformed. The piston is given a small displacement x towards left. Then



(a) the pressure in left chamber increases

(b) the pressure in right chamber decreases

(c) the piston oscillates with time period $2\pi\sqrt{\frac{Ml_0}{2PA+l_0K}}$

(d) the piston oscillates with time period $2\pi\sqrt{\frac{2Ml_0}{PA+l_0K}}$

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10. A charge particle of charge q, mass m is projected with velocity v in a region of electric and magnetic fields E and B respectively, such that v, E and B are mutually perpendicular to each other and $v = E/B$. Then (neglect gravity)

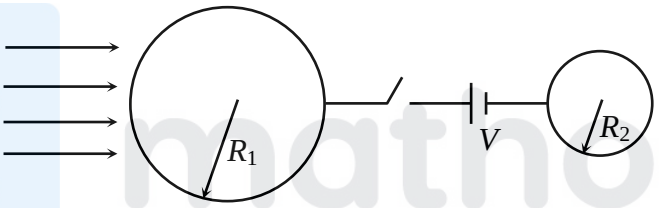
(a) If E is switched off, then particle will follow a circular path

(b) If B is switched off, then particle will follow a parabolic path

(c) If E is switched off, for a time of $\frac{\pi m}{4qB}$ and then switched on, the path of particle will not be a straight line

(d) If E is switched off, for a time of $\frac{2\pi m}{qB}$ and then switched on, the path of particle will be a straight line

11. Two conducting uncharged spheres of radius R_1 and R_2 ($R_1 > R_3$) are connected to a battery with a switch as shown. Light rays of frequency of f are incident on the bigger sphere and simultaneously the switch is closed. Work function of bigger sphere is ϕ. After some time the charge on bigger sphere becomes q_1, and on smaller sphere becomes $-q_2$ remains constant there after. Then



(a) $\frac{q_1}{4\pi\varepsilon_0 R_1} + \frac{q_2}{4\pi\varepsilon_0 R_2} = V$

(b) The number of electrons emitted by larger sphere is $\frac{q_1 - q_2}{e}$ when e is the charge one electron

(c) $\frac{eq_1}{4\pi\varepsilon_0 R_1} = hf - \phi$

(d) The number of electrons emitted by larger sphere is independent of the potential of the battery

SECTION – III

Reasoning Type

This section contains 5 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

12. **Statement-1**: Two blocks A and B of mass M and $2M$ respectively are connected by a massless spring of force constant K. They are stretched by an amount of x and then released on a smooth horizontal table. The relative velocity of the blocks when spring comes to its natural length is $x\sqrt{\frac{3K}{2M}}$.

Statement-2: When there is no external force acting on a system, its momentum and mechanical energy is always conserved.

(a) A (b) B (c) C (d) D

13. **Statement-1**: The time period of simple oscillation of a mass supported by a spring is T. The spring is cut into 4 equal parts and the body is again suspended by one of the pieces. The new time period of simple oscillation will be $\frac{T}{2}$.

Statement-2: The spring constant of a spring is inversely proportional to the length of the spring.

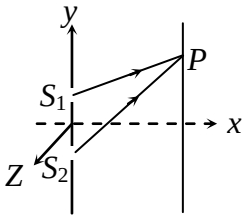
(a) A (b) B (c) C (d) D

14. **Statement-1**: If in an A.C. circuit, the applied voltage is in phase with the current flowing in the circuit, then the circuit may have only resistance in it.

Statement-2: If there is no phase difference between current and applied voltage, the power factor of the circuit is unity.

(a) A (b) B (c) C (d) D

15. **Statement-1**: In a young's double slit experiment set up shown in the figure, screen is perpendicular to x-axis. The fringes will be hyperbolic in shape and central bright fringe will be a straight line.



Statement-2: Each fringe will be the locus of a point for which $S_2P - S_1P$ will be a constant

(a) A (b) B (c) C (d) D

16. **Statement-1**: Two particles are projected towards each other from a very large distance along the line joining them on a smooth frictionless surface. Their speeds may decrease as they approach each other.

Statement-2: The net force on the system of the above two particles will be zero.

(a) A (b) B (c) C (d) D

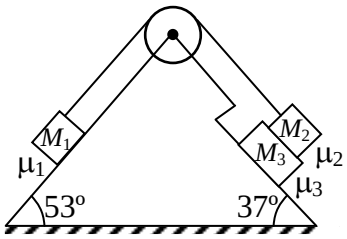
SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

	1	2	3	4	5
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

1. In the given arrangement there are three boxes with masses M_1, M_2 and M_3, 'μ_1, μ_2, μ_3' are the co-efficient of friction between M_1 and inclined plane, between M_2 & M_3, between M_3 & inclined plane respectively. Column I gives the values of parameter $M_1, M_2, M_3, \mu_1, \mu_2$ and μ_3 respectively and column II gives certain aspects of the system corresponding to the values given in column I.



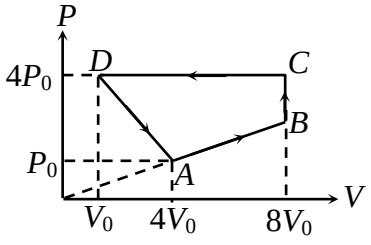
f_1 = friction force between M_1 and inclined plane

f_2 = friction force between M_2 & M_3

f_3 = friction force between M_3 and inclined plane (sin 37^0 – 3/5, sin 53^0 = 4/5)

Column I	Column II
(A) M_1 = 5kg, M_2 = 3kg, M_3 = 2kg, $\mu_1 = \mu_2 = \mu_3 = 0$	**1.** acceleration of M_1 and M_2 is same
(B) M_1 = 5kg, M_2 = 3kg, M_3 = 2kg, $\mu_1 = \mu_3 = 0, \mu_2 = 3/4$	**2.** acceleration of M_2 and M_3 is same
(C) $M_1 = M_2$ = 4kg, M_3 = 2kg $\mu_1 = 1/2, \mu_2 = 3/4, \mu_3 = 0$	**3.** $f_1 \geq f_2$
(D) $M_1 = M_2$ = 4kg M_3 = 2kg $\mu_1 = \mu_3 = 1/2, \mu_2 = 3/4$	**4.** $f_1 < f_2$
	5. $f_2 \leq f_3$

2. $\frac{3}{25}$ moles of an ideal monatomic gas undergoes a process as shown in figure. ΔU, Q and W represents the change in internal energy, amount of heat exchange and pressure-volume work respectively for the gas either for single process or for complete cycle. ($R = \frac{25}{3}$ SI unit)

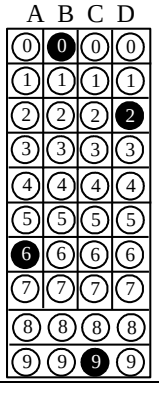


Column I		Column II	
(A)	In process AB	**1.**	$Q = 0$ or Q = +ve
(B)	In process DA	**2.**	$\Delta U = 0$ or ΔU = –ve
(C)	In cyclic process $ABCD$	**3.**	ΔU = +ve
(D)	In process CD	**4.**	W = +ve or $W = 0$
		5.	W = –ve

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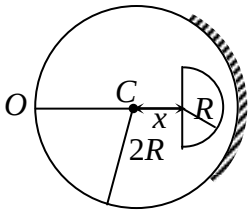
SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



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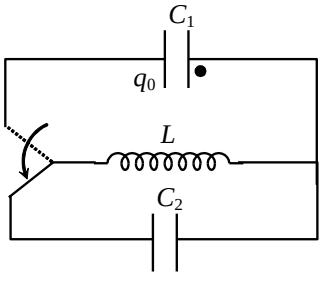
1. A hemispherical cavity of radius R is carved out from a sphere (μ = 1.5) of radius $2R$ such that principal axis of cavity coincides with that of sphere. One side of sphere is silvered as show in figure. Find the value of x (in cm) for which the image of an object at O is formed at O it self. (Take R = 8 cm and $\sqrt{145}$ =12)



Space for work

mathongo

2. At time $t = 0$ a charge q_0 is given at the plate of capacitor C_1 (C_1 = 400 μF) forming an LC circuit with inductor L = 100 H. At $t = \frac{\pi}{10}$ seconds connection with the capacitor C_1 is broken and another capacitor C_2 is joined with the inductor. At time $t = \frac{11n\pi}{200}$ the energy stored in capacitor C_2 is 36% of the total energy available finally. If C_2 = 100 μF find the value of n, $\left(\sin\left(\frac{\pi}{10}\right) = \frac{3}{10}\right)$



Space for work

mathongo

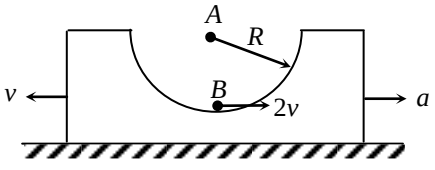
3. A non-conducting uniformly charged sphere having charge ‘96 columbs’ and radius ‘10m’ is placed with its center at origin, at a distance of ‘5m’ from the origin there is a large imaginary sheet placed, the flux of electric field through the intersection area of the sheet and the sphere is '$x/\in_0$'. Calculate the value of x?

Space for work

mathongo

4. A bead kept at the bottom of a wedge moves towards right with a velocity $2v$. If the wedge moves towards left with a velocity v while accelerating towards right with an acceleration a as shown then the magnitude of acceleration of the bead is x units. Find x if

$v = 3$ and $R = 9$ units.



Space for work

mathongo

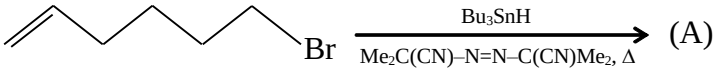
CHEMISTRY

SECTION – I

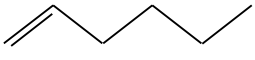
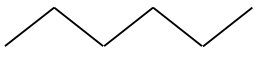
Single Correct Choice Type

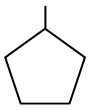
This section contains 4 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. 

The major product (A) of the given reaction would be

(a)  (b) 

(c)  (d) 

2. The rate of a certain reaction depends on concentration according to the equation:

$$-\frac{d[A]}{dt} = \frac{k_1[A]}{1+k_2[A]} \quad \text{where } k_1 > k_2$$

Which of the following statement is not correct for the above reaction?

(a) At low concentration of A, ($[A] << k_2$) the order of reaction is unity.

(b) At high concentration of A ($[A] >> k_2$), order of reaction is two.

(c) At low concentration of A ($[A] << k_2$), $-\frac{d[A]}{dt} \simeq k_1[A]$.

(d) At high concentration of A ($[A] >> k_2$), order of reaction is zero.

3. A bulb of capacity 1 dm^3 contains 1.004×10^{23} gaseous hydrogen molecules and the pressure exerted by these molecules is 101.325 kP_a, the root mean square speed of hydrogen molecules is ($N_A = 6.023 \times 10^{23}$, R = 0.821 L atm mol^{-1} K^{-1} = 8.314 J K^{-1} mol^{-1})

(a) $\sqrt{9.1\times10^5}$ ms^{-1} (b) $\sqrt{3.23\times10^5}$ ms^{-1}

(c) $\sqrt{9.1\times10^4}$ ms^{-1} (d) $\sqrt{3.23\times10^4}$ ms^{-1}

4. Consider the following changes:

1. $M(s) \longrightarrow M(g)$
2. $M(s) \longrightarrow M^{2+}(g) + 2e^-$
3. $M(g) \longrightarrow M^+(g) + e^-$
4. $M^+(g) \longrightarrow M^{2+}(g) + e^-$
5. $M(g) \longrightarrow M^{2+}(g) + 2e^-$

The second ionization energy of M could be calculated from the energy values associated with

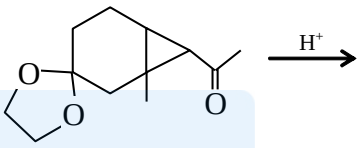
(a) 1 + 5 (b) 2 + 4
(c) 5 – 3 (d) 2 – 1 + 3

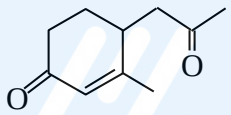
SECTION – II
Multiple Correct Choice Type

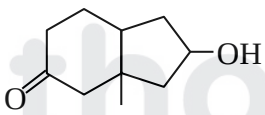
This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

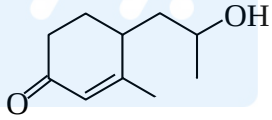
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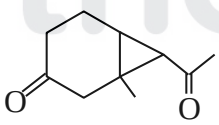
5. Identify the product(s) formed during the following reaction.



(a) 

(b) 

(c) 

(d) 

6. 10 mL of a weak acid (HA) solution is 20% dissociated in water. This solution is completely neutralized by 10 mL of 2×10^{-3} M NaOH solution. Which of the following is/are true about the given case? (log 2 = 0.3)

(a) pH of weak acid (HA) solution was 3.4.
(b) The dissociation constant of weak acid (HA) was 1×10^{-4}.
(c) pH of the solution after addition of NaOH is 7.5.
(d) K_h (hydrolysis constant) of the salt is 1×10^{-9} M.

7. Which of the following statement(s) is/are correct about acetone and trichloromethane mixture?

(a) Mixtures of acetone and trichloromethane show positive deviation from Raoult's law.

(b) The force of attraction acting between molecules of acetone and trichloromethane in a mixture are greater than those acting between the molecules in pure acetone.

(c) Pure acetone can be obtained by the careful fractional distillation of any mixture of acetone and trichloromethane.

(d) When acetone and trichloromethane are mixed, the enthalpy change is negative.

8. For the given reaction, $X_2 + H_2O \longrightarrow X^- + HOX + H^+$

Which of the following statement(s) is/are true?

(a) It is a nucleophilic displacement reaction.

(b) It is a second order reaction.

(c) The unit of rate constant of the reaction is sec^{-1}.

(d) The HOX is a linear molecule.

SECTION – III
Reasoning Type

This section contains 4 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True ; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True ; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

9. **Statement-1:** Aqueous ferrous sulphate is a very good absorber for NO, the compound formed in this reaction is found to contain three unpaired electrons.

Statement-2: Paramagnetic species will always contain odd number of electrons.

(a) A (b) B (c) C (d) D

10. **Statement-1:** Mixture of gases containing one mole each of monoatomic gas and diatomic gas gives value of C_p/C_v as 1.5.

Statement-2: Value of C_p/C_v is independent of moles of the gases taken.

(a) A (b) B (c) C (d) D

11. **Statement-1:** When malonic acid is heated alone, it undergoes decarboxylation whereas heating it with P_2O_5 results in dehydration.

Statement-2: Dicarboxylic acids on heating alone undergo decarboxylation only.

(a) A (b) B (c) C (d) D

12. **Statement-1:** At very high pressure, value of compressibility factor (Z), for all gases is greater than one.

Statement-2: At very high pressure, forces of repulsions dominate over attractive forces.

(a) A (b) B (c) C (d) D

SECTION–IV
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

On dissolving a sparingly soluble salt in water, after some time an equilibrium is established when the rate of dissolution of ions from the solid equals the rate of precipitation of ions from the saturated solution at a particular temperature. For example, when solid AgCl is added to water, following equilibrium is found to exist.

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$

$$K_{eq} = \frac{[Ag^+][Cl^-]}{[AgCl]}$$

$$K_{sp} = K_{eq}[AgCl] = [Ag^+][Cl^-]$$

$K_{eq}[AgCl]$ is known as solubility product (K_{sp}). The principle of solubility product applies to all solution of slightly soluble materials whether they dissociate in one step or in several steps.

13. The salt $Al(OH)_3$ is involved in the following two equilibria:

$Al(OH)_3(s) \rightleftharpoons Al^{3+}(aq) + 3OH^-(aq)$; K_{sp}

$Al(OH)_3(s) + OH^-(aq) \rightleftharpoons Al(OH)_4^-(aq)$; K_c

Which of the following relationship is correct at the point of minimum solubility of $Al(OH)_3$?

(a) $[OH^-] = \left(\frac{K_{sp}}{K_c}\right)^{1/3}$ (b) $[OH^-] = \left(\frac{K_c}{K_{sp}}\right)^{1/4}$

(c) $[OH^-] = \sqrt{\left(\frac{K_{sp}}{K_c}\right)^{1/4}}$ (d) None of these

14. Consider the following statement and select the correct option.

(I) The concentration of Fe^{3+} will increase in a solution of $Fe(OH)_3$ when $[H^+]$ decrease. (Given: K_{sp} of $Fe(OH)_3$ is 4×10^{-38} at 298 K)

(II) In a mixture of NH_4Cl and NH_4OH in water, a further amount of NH_4Cl is added the pH of mixture will decrease.

(III) An aqueous solution of NH_4I and HCO_2K will be basic and acidic respectively.

(a) only (I) is correct. (b) only (II) is correct.

(c) only (III) is correct. (d) (II) and (III) are correct.

15. The simultaneous solubility of $AgCN(K_{sp} = 2.5 \times 10^{-16})$ and $AgCl$ $(K_{sp} = 1.6 \times 10^{-10})$ in 1M $NH_3(aq)$ are $\{\text{Given: } K_f[Ag(NH_3)_2^+]\} = 10^7$.

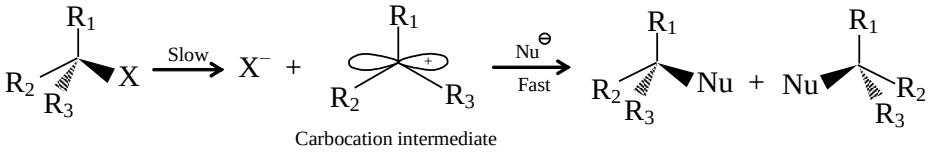
(a) 0.037 & 5.78×10^{-8} respectively (b) 5.78×10^{-8} & 0.037 respectively

(c) 0.04 & 6.25×10^{-8} respectively (d) 1.58×10^{-3} & 1.26×10^{-5} respectively

Passage-II

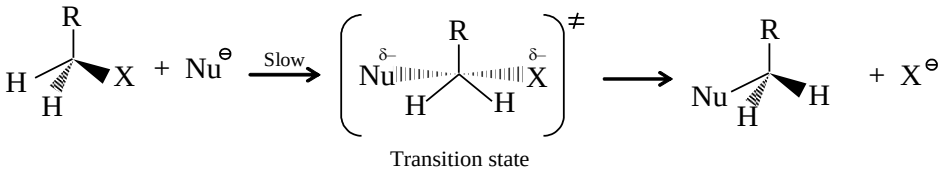
Nucleophilic substitution in alkyl halides usually follows either of the two pathways.

S_N1 process:



This pathway is preferred for tertiary halides.

S_N2 process:



Primary alkyl halides preferably follows S_N2 pathway. Secondary alkyl halides follows dual pathway. Some part of it goes via S_N1 and remaining part of it goes through S_N2 process.

Optically active (S)–(+)–2–bromo octane, $[\propto]_D^{25^o}$ = +36°, reacts with aqueous NaOH in acetone to give optically pure (R)–(–)–2–octanol, $[\propto]_D^{25^o}$ = –10°. With partially racemized bromo compound whose $[\propto]_D^{25^o}$ = +30°, the $[\propto]_D^{25^o}$ of the alcohol product is –6°.

16. The optical purity of partially racemized alcohol produced is approximately
(a) 83% (b) 60% (c) 17% (d) 40%

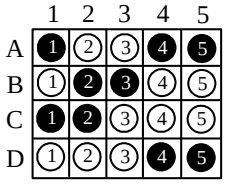
17. The percentage of inversion observed in the reaction is
(a) 66% (b) 28% (c) 72% (d) 34%

18. The percentage of reaction involving front side attack on the bromo compound is
(a) 14% (b) 17% (c) 83% (d) 86%

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

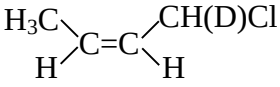
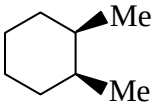
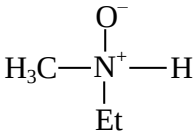
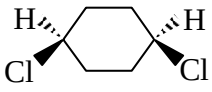
If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

Column I	Column II
(A) $[Cr(gly)_3]$	**1.** Linkage isomerism.
(B) $[CoBr_2Cl_2(SCN)_2]^{4-}$	**2.** Spin only magnetic moment = $\sqrt{15}$.
(C) $[Fe(NH_3)_6]Cl_3$	**3.** Optical isomerism.
(D) $K[PtBrCl(NO_2)(NH_3)]$	**4.** Geometrical isomerism. **5.** Octahedral geometry.

2.

Column I	Column II
(A) 	**1.** Optically active
(B) 	**2.** Optically inactive
(C) 	**3.** Plane of symmetry
(D) 	**4.** Number of chiral center is 1. **5.** Number of stereocenter is 2.

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MATHEMATICS

SECTION – I
Single Correct Choice Type

This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. Variable plane $lx + my + nz = p$ (normal form with p constant) intersects the co-ordinate axes at A, B, and C respectively, then the least area of ΔABC, is

(a) $\frac{27}{4}p^2$ (b) $\frac{3\sqrt{3}}{2}p^2$

(c) $\frac{p^2}{3\sqrt{3}}$ (d) none of these

2. Two regular polygons are inscribed in same circle. The first polygon has 1982 sides and second has 2973 sides. If the polygons have any common vertices, the number of such vertices are

(a) 1 (b) 990 (c) 991 (d) none of these

3. The curve represented by the equation $\frac{x^2}{\sin\sqrt{2} - \cos\sqrt{3}} + \frac{y^2}{\sin\sqrt{3} - \cos\sqrt{2}} = 1$ is

(a) an ellipse with foci on x-axis
(b) an ellipse with foci on y-axis
(c) an hyperbola with foci on x-axis
(d) an hyperbola with foci on y-axis

4. The value of $\sum_{r=1}^{n} \frac{r}{4r^4 + 1}$ is equal to

(a) $\frac{\sum n}{4\sum n + 3}$ (b) $\frac{\sum n}{2\sum n + 3}$ (c) $\frac{\sum n}{8\sum n + 3}$ (d) $\frac{\sum n}{4\sum n + 1}$

5. The least value of k such that the equation $(\ln x) + k = e^{x-k}$ has a solution is

(a) e (b) $1/e$ (c) 1 (d) none of these

6. Let A, B and C be square matrices of order 3×3 with real elements. If A is invertible and $(A - B)C = BA^{-1}$, then

(a) $C(A-B)=A^{-1}B$ (b) $C(A-B)=BA^{-1}$

(c) $(A-B)C=A^{-1}B$ (d) all the above

7. Given two real sets $A=\{a_1, a_2, ...a_{2n}\}$ and $B\{b_1, b_2, ...b_n\}$. If $f:A\to B$ is a function such that every element of B has pre-image and $f(a_1)\le f(a_2)\le .....\le f(a_{2n})$, then number of such mappings are

(a) ${}^{2n}C_n$ (b) ${}^{2n}C_{n-1}$

(c) ${}^{2n-1}C_{n-1}$ (d) none of these

8. If $x_1=\sqrt{3}$ and $x_{n+1}=\dfrac{x_n}{1+\sqrt{1+x_n^2}}$, $\forall n\in N$, then $\lim\limits_{n\to\infty}2^n x_n$ equals to

(a) $\dfrac{3}{2\pi}$ (b) $\dfrac{2}{3\pi}$

(c) $\dfrac{2\pi}{3}$ (d) $\dfrac{3\pi}{2}$

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

9. Let $S_n=\lim\limits_{n\to\infty}\left(\dfrac{1}{n^6}+\dfrac{32}{n^6}+.....+\dfrac{1}{n}\right)$ and $T_n=\lim\limits_{n\to\infty}\left(\dfrac{1}{n^6}+\dfrac{32}{n^6}+.....+\dfrac{(n-1)^5}{n^6}\right)$, then which of the following is/ are true?

(a) $S_n\to\dfrac{1}{6}^+$ (b) $(S_n+T_n)<\dfrac{1}{3}$

(c) $(S_n+T_n)>\dfrac{1}{3}$ (d) $T_n\to\dfrac{1}{6}^-$

10. $z_1, z_2.....z_n$ be non-zero complex numbers of equal modulus and satisfying the equation $z^n-|z|z^{n-1}+1=0$, then

(a) $\text{Re}\left(\sum_{j=1}^{n}\sum_{k=1}^{n}\frac{z_j}{z_k}\right)=0$ (b) $\text{Re}\left(\sum_{j=1}^{n}\sum_{k=1}^{n}\frac{z_j}{z_k}\right)=1$

(c) $\text{Im}\left(\sum_{j=1}^{n}\sum_{k=1}^{n}\frac{z_j}{z_k}\right)=0$ (d) $\text{Im}\left(\sum_{j=1}^{n}\sum_{k=1}^{n}\frac{z_j}{z_k}\right)=1$

11. $f:[0,1]\to R$ is a differentiable function such that $f(0)=0$ and $|f'(x)|\le k|f(x)|$ for all $x \in [0,1]$, $(k>0)$, then which of the following is/are always true?

(a) $f(x)=0,\ \forall x\in R$ (b) $f(x)=0,\ \forall x\in[0,1]$

(c) $f(x)\neq 0,\ \forall x\in[0,1]$ (d) $f(1)=k$

12. If $\log_2\left(\log_{\frac{1}{2}}(\log_2(x))\right)=\log_3\left(\log_{\frac{1}{3}}(\log_3(y))\right)=\log_5\left(\log_{\frac{1}{5}}(\log_5(z))\right)=0$ for positive x, y and z, then which of the following is/ are NOT true?

(a) $z<x<y$ (b) $x<y<z$

(c) $y<z<x$ (d) $z<y<x$

SECTION–III
Comprehension Type

This section contains 1 group of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Consider a convex quadrilateral *ABCD* with diagonals intersecting at *P*. Ellipse E_1 has foci *A* and *B* and is passing through *P*, ellipse E_2 has foci *C* and *D* and is also passing through *P*. Both E_1 and E_2 touch each other at *P*. This can be proved by contradiction assuming that E_1 and E_2 intersect at *P* and *Q* therefore by property of ellipse $PA+PB=QA+QB$, $PC+PD=QC+QD$. But if *Q* is not the point of intersection of diagonals, then $QA+QC>AC$ (triangle inequality) therefore *P* and *Q* are same points.

13. Ellipse E_3 has foci *B* and *C* and passes through *P*, then which of the following is incorrect?

(a) E_3 is orthogonal to E_1 at *P* (b) E_3 is orthogonal to E_2 at *P*

(c) circumcircle of BCP touches E_3 at P (d) E_3 has eccentricity $\frac{BC}{(BP+PC)}$

14. If $ABCD$ is a cyclic quadrilateral and eccentricities of E_1, E_2 and E_3 are e_1, e_2 and e_3 respectively, then

(a) $e_1 = e_2 = e_3$ (b) $e_1 = e_2$

(c) $e_3 = e_1$ (d) $e_3 = e_2$

15. If the angle bisector of $\angle BPA$ intersects AB and CD at M and N respectively, then $\frac{(MB+CN)}{BC}$ is equal to

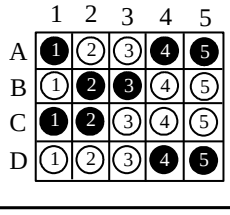
(a) $\frac{e_1}{e_3}$ (b) $\frac{e_3}{e_1}$

(c) $e_1 e_3$ (d) $\frac{1}{e_1 e_3}$

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

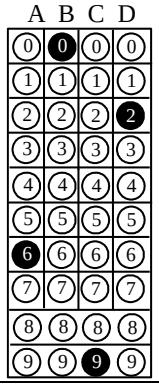
Column I	Column II
(A) $f(x)$ is defined for $x \geq 0$ and has a continuous derivative such that $f(0)=1;\ f'(0)=0$ and $(1+f(x))f''(x)=1+x$, then $f(x)$ is a/an	**1.** cubic polynomial function
(B) If $f(x).f(y)=f(x+y),\ \forall x,\ y \in R$ and $f(0)\neq 0,\ f(1)=e,\ f'(0)=1$, then $f(x)$ is a/an	**2.** increasing function
(C) If $f(x)$ is polynomial function satisfying $f(0)=0$ and $f(x^2+1)=f^2(x)+1$, then $f(x)$ is a/an	**3.** periodic function
(D) If f is a polynomial function such that $f: R^+ \to R^+$ and $f(f(x))=6x-f(x)$, then $f(x)$ is a/an	**4.** linear polynomial function **5.** quadratic polynomial function

2. Triangle is formed by joining the vertices of a cube.

	Column I	Column II
(A)	Probability that the triangle is isosceles (including equilateral) is	**1.** $\frac{3}{7}$
(B)	Probability that the triangle is isosceles (excluding equilateral) is	**2.** $\frac{4}{7}$
(C)	Probability that the triangle is scalene is	**3.** $\frac{6}{7}$
(D)	Probability that the triangle is right angled triangle is	**4.** $\frac{3}{14}$ **5.** $\frac{5}{14}$

SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



1. Let $p(x) = x^5 + x^2 + 1$ have roots x_1, x_2, x_3, x_4 and x_5 , $g(x) = x^2 - 2$, then find the value of $g(x_1)g(x_2)g(x_3)g(x_4)g(x_5) - 30g(x_1x_2x_3x_4x_5)$.

Space for work

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2. If $\alpha = e^{i2\pi/7}$ and $f(x) = A_0 + \sum_{k=1}^{20} A_k x^k$ and the value of

$f(x) + f(\alpha x) + f(\alpha^2 x) + ... + f(\alpha^6 x)$ is $k(A_0 + A_7 x^7 + A_{14} x^{14})$, then find the value of k.

Space for work

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3. $f : R \to R$ is $f(x) = \ln(x + \sqrt{x^2 + 1})$, then find the number of solutions to the equation $\left| f^{-1}(x) \right| = e^{-|x|}$.

Space for work

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4. $A(0, 2)$, B and C are points on parabola $y^2 = x + 4$ and such that $\angle CBA = \frac{\pi}{2}$, then find the least positive value of ordinate of C.

Space for work

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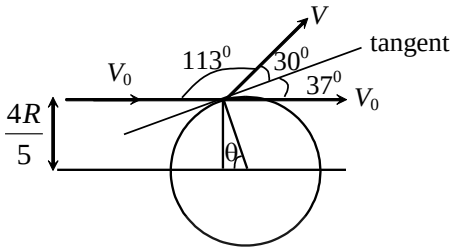
PAPER-I
PHYSICS
SECTION – I
Single Correct Choice Type

1. $\dfrac{1}{R_{eq.}} = \int \dfrac{1}{dR} = \int_{R_1}^{R_2} \dfrac{2Ldx}{\rho\pi x} = \dfrac{2L}{\pi\rho}\ln\dfrac{R_2}{R_1}$

$\therefore \quad R_{eq} = \dfrac{\pi\rho}{2L\ln\left(\dfrac{R_2}{R_1}\right)}$

$\therefore$ **(c)**

2. Since the surfaces are smooth so tangential component of velocity would be conserved.



$$\sin\theta = \frac{4R}{5\times R} = \frac{4}{5}$$

$\theta = 53^0$

So $V_0\cos 37^0 = V\cos 30^0$

or $V_0 \times \dfrac{4}{5} = \dfrac{\sqrt{3}}{2}V$

$8V_0 = 5\sqrt{3}V$

$\therefore$ **(c)**

3. $\sum F_{\text{ext}} = 0$ on 'board and man' system

$$m\times\frac{5l}{2} + m\times 2l = (m+m)x$$

or $x = \dfrac{9l}{4}$

$\therefore$ **(d)**

4. For pure rolling motion

$\alpha.R - 6 = 3$

or $\alpha R = 6 + 3$

$\alpha = \dfrac{9}{3} = 3\,\text{rad/s}^2$

$\therefore$ **(a)**

5. Net charge flowing through switch A is $60\times 3 - 60\times 2 = 60\mu\,\text{C}$

$\therefore$ **(d)**

6. $2A\sin kx = 3\sqrt{2}$

$2\times 3\sin kx = 3\sqrt{2}$

$\sin kx = \dfrac{1}{\sqrt{2}}$

$\dfrac{2\pi}{\lambda}x = \dfrac{\pi}{4};\ \dfrac{3\pi}{4}$

$x = \dfrac{\lambda}{8};\dfrac{3\lambda}{8}$ .....

Distance between consecutive points = $\dfrac{3\pi}{8}-\dfrac{\lambda}{8}=\dfrac{\lambda}{4}$

$\dfrac{\lambda}{4} = 20\,\text{cm}$

$\Rightarrow$ $\lambda = 80$ cm

So, $(n+1)\dfrac{\lambda}{2} = 240$

$\Rightarrow$ $(n+1)\dfrac{80}{2} = 240$

or $n+1 = 6$

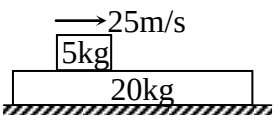
$n = 5$

So string is vibrating in fifth overtone

$\therefore$ **(d)**

SECTION – II
Multiple Correct Choice Type

7. Conserving the momentum



$5\times 25 = 25\times v$

$\Rightarrow$ $v = 5$ m/s

$\Delta K + \Delta U = W_{\text{friction}}$

$\Rightarrow$ $\dfrac{1}{2}(5+20)5^2 - \dfrac{1}{2}\times 5\times 25^2 + 0 = W_f$

$= \dfrac{1}{2}\times 625\times(-4) = -1250\,\text{J}$

$W_{\text{friction}} = -\mu mg\ \ d_{\text{rel}} = -1250$ J

$\Rightarrow$ $\dfrac{5}{8}\times 5\times 10\times d_{rel} = -1250$

$d_{rel} = 40\,\text{m}$

Now, $d_{\text{rel}} = \dfrac{1}{2}a_{rel}.t^2$

$\Rightarrow$ $40 = \dfrac{1}{2}\left(\mu g + \dfrac{\mu mg}{M}\right)t^2$

or $40 = \frac{1}{2} \times 15.625 \times t^2$

$\Rightarrow \quad t = \frac{16}{5} \text{sec}$

$\therefore$ **(a, b, d)**

8. Acceleration of the whole system towards right:

$$a = \frac{F}{M+m}$$

$$F - mg\sin\theta = ma\cos\theta$$

$\Rightarrow \quad F - mg\sin\theta = m\frac{F}{M+m}\cos\theta$

$\Rightarrow \quad F = \frac{(M+m)mg\sin\theta}{M+m-m\cos\theta}$

Pseudo force on m as seen from the frame of M:

$$F_{s1} = ma = \frac{mF}{m+M}$$

$$= mg\sin\theta\left(\frac{m}{M+m(1-\cos\theta)}\right) < mg\sin\theta$$

Pseudo force on M as seen from the frame of m:

$$F_{s2} = Ma = \frac{MF}{m+M}\left(> \frac{mF}{m+M}\right)$$

$$= mg\sin\theta\left(\frac{M}{M+m(1-\cos\theta)}\right) < mg\sin\theta$$

Now $mg\cos\theta - N = ma\sin\theta$

$\Rightarrow \quad N = mg\cos\theta - ma\sin\theta$

Hence N is less than $mg\cos\theta$. Hence it will also be less than $mg\sin\theta$, because $\theta = 45°$.

Applying equation on m in horizontal direction:

$$F\cos\theta - N\sin\theta = ma$$

$\Rightarrow \quad F\cos\theta - N\sin\theta = m\frac{F}{M+m}$

$\Rightarrow \quad N = \frac{mF}{M+m}\left(\frac{(M+m)\cos\theta - m}{m\sin\theta}\right)$

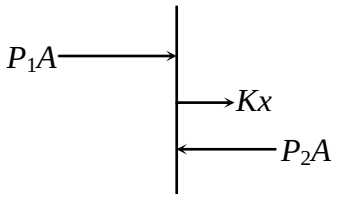
Put $\theta = 45°$

$\Rightarrow \quad N = \frac{mF}{M+m}\left(\frac{M+m-\sqrt{2}m}{m}\right) > \frac{mF}{m+M}$

Normal force between ground and M will be $(M + m)\,g$. It is greater than $mg\sin\theta$. It is also greater than $\frac{mF}{M+m}$, because $\frac{mF}{M+m}$ is less than $mg\sin\theta$.

$\therefore$ **(a, b, c)**

9. The FBD of the piston is



Restoring force = $(P_2 - P_1)A + Kx$

$$-Ma = (P_2 - P_1)A + Kx$$

Also $P_1A(l_0 - x) = PAl_0$

$$P_1 = \frac{Pl_0}{l_0 - x} \quad \ldots \text{(i)}$$

$$P_2A(l_0 + x) = PAl_0$$

$$P_2 = \frac{Pl_0}{l_0 + x} \quad \ldots \text{(ii)}$$

From (i) and (ii) $a = \left[\frac{2Pl_0A}{l_0^2 - x^2} + K\right]\frac{x}{M}$

Since $x << l_0$

$$a = -\left[\frac{2Pl_0A}{l_0^2} + K\right]\frac{x}{M}$$

$\therefore \quad T = 2\pi\sqrt{\frac{Ml_0}{2PA + Kl_0}}$

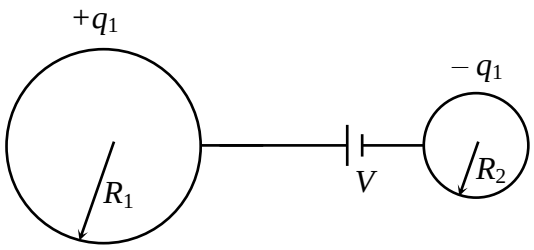
$\therefore$ **(a, b, c)**

10. If only magnetic field is present, the particle will follow a circular path and if only electric field is present, particle will follow a parabolic path.

$\therefore$ **(a, b, c, d)**

11. $\frac{q_1}{4\pi\varepsilon_0 R_1} - \left(-\frac{q_2}{4\pi\varepsilon_0 R_2}\right) = V$

Also number of electrons emitted = $\frac{q_1 - q_2}{e}$



$\therefore$ **(a, b, c, d)**

SECTION–III
Reasoning Type

12. $v_{rel} = \vec{v}_1 - \vec{v}_2$; $\vec{v}_1 = x\sqrt{\frac{K}{M}}$; $\vec{v}_2 = x\sqrt{\frac{K}{2M}}$

$\therefore$ **(c)**

13. $K \propto \frac{1}{l}$ where l is the length of the spring

$\therefore \quad K' = 4K$

$\therefore$ **(a)**

14. $\tan\phi = \dfrac{X_L - X_C}{R}$

If $\phi = 0 \Rightarrow X_L = X_C$

$\therefore \quad Z = R$

$\therefore$ **(b)**

15. The path difference at central point will be zero.

For other fringes, $S_2P - S_1P = n\lambda$ (for constructive interference)

$\therefore$ **(a)**

16. If two particles approach each other under mutual forces, the net force on the system is zero. However, the speed may increase or decrease depending upon the nature of force.

$\therefore$ **(b)**

SECTION–IV
Matrix–Match Type

1. Blocks of mass M_1 and M_2 are connected with a string. Hence their acceleration must be equal.

If the friction force between M_2 and M_3 is sufficient to prevent relative slipping, they move with same acceleration.

$\therefore$ **(A) – (1, 3, 5) ; (B) – (1, 2, 4) ; (C) – (1, 2, 4; D) – (1, 2, 3, 5)**

2. In process *AB*, $P \propto V$.

Also along *AB*, temperature of the gas increases

In process *BC*, volume is constant and pressure increases. Hence temperature will increase.

In process *CD*, pressure is constant and volume decreases. Hence temperature will decrease.

In process *DA*, pressure decreases and volume increases

Also the process *DA* is of the form $P = -mV + C$ where m and C are constants.

Using ideal gas equation

$$\frac{nRT}{V} = -mV + C$$

$$T = -\frac{m}{nR}V^2 + \frac{C}{nR}V$$

Hence temperature first increases then decreases

$\therefore$ **(A) – (1, 3, 4) ; (B) – (1, 2, 4); (C) –2, 5; (D) – 2, 5**

SECTION–V
Integer Answer Type

1. Refraction at plane surface

$u = -(2R + x)$

$$\frac{1}{v} - \frac{\mu}{-(2R+x)} = 0$$

$$v = -\frac{(2R+x)}{\mu}$$

For refraction at curved surface

$$u = -\left(R + \frac{2R+x}{\mu}\right)$$

$$\frac{\mu}{v'} - \frac{1}{-\left(R + \frac{2R+x}{\mu}\right)} = \frac{\mu - 1}{-R}$$

$$v' = \frac{-\mu}{\frac{\mu-1}{R} + \frac{\mu}{2R + R\mu + x}} \qquad \ldots \text{(i)}$$

Now in order to form the final image on O itself this ray should seen to be coming from centre of curvature C.

So $v' = -(R + x)$ … (ii)

From (i) and (ii) and substituting $\mu = 3/2$

$$2x^2 + 9Rx - 8R^2 = 0$$

$$\Rightarrow \quad x = \frac{-9R + \sqrt{145R^2}}{4} = \frac{3R}{4}$$

$x = 6$ cm

Ans.: 6

2. Initially $\omega = \frac{1}{\sqrt{LC_1}} = \frac{1}{\sqrt{400 \times 10^{-6} \times 100}} = 5$ rad/s

$$q = q_0 \cos(\omega t)$$

$$\Rightarrow \quad q = q_0 \cos\left(\frac{\pi}{10} \times 5\right) = 0$$

So at $t = \frac{\pi}{10}$ sec; all the energy of capacitor C_1 would be stored in inductor.

Now, $\omega = \sqrt{\frac{1}{LC_2}} = \sqrt{\frac{1}{100 \times 100 \times 10^{-6}}}$ = 10 rad/s

36% of $\frac{q_0^2}{2C_1} = \frac{q^2}{2C_2}$

$q = \frac{3}{10} q_0$

$\Rightarrow \quad \frac{3q_0}{10} = q_0 \sin(10t)$

$\Rightarrow \quad \sin(10t) = 0.3 = \sin\frac{\pi}{10}$

$t = \frac{\pi}{100}$

Total time $= \frac{\pi}{10} + \frac{\pi}{100} = \frac{11\pi}{100} = \frac{11n\pi}{200}$

So $n = 2$

Ans.: 2

3. Flux of electric field through the intersection area is given by

$\oint \vec{E}.\overrightarrow{ds} = \frac{Qr}{4 \in_0 R^3}(R^2 - r^2)$ where r is distance of sheet from the origin

$= \frac{Q}{8 \in_0 R^2}\left(R^2 - \frac{R^2}{4}\right) = \frac{3Q}{32 \in_0} = \frac{9}{\in_0}$

$x = 9$

Ans.: 9

4. As the wedge accelerates with $\vec{a} = a\hat{i}$ the centre A also has same acceleration.

$\therefore \quad \vec{a}_A = a\hat{i}$

$\vec{v}_A = -v\hat{i}$

$\vec{a}_{B,g} = \vec{a}_{B,A} + \vec{a}_A$ where $\vec{a}_{B,A} = (\vec{a}_{B,A})_t + (\vec{a}_{B,A})_r$ and $\vec{a}_A = a\hat{i}$

$\therefore \quad \vec{a}_{B,g} = \vec{a}_B = (\vec{a}_{B,A})_t + (\vec{a}_{B,A})_r + a\hat{i}$

$(\vec{a}_{BA})_t = \vec{a}_{B_x} - \vec{a}_{A_x} = 0 - a\hat{i}$

$\vec{v}_{B,A} = \vec{v}_B - \vec{v}_A = 2v\hat{i} - (-v\hat{i}) = 3v\hat{i}$

$\therefore \quad (\vec{a}_{B,A})_r = \frac{9v^2}{R}\hat{j}$

$\therefore \quad \vec{a}_{B,g} = -a\hat{i} + \frac{9v^2}{R}\hat{j} + a\hat{i} = \frac{9v^2}{R}\hat{j}$

Putting $v = 3$ and $R = 9$

We get $|\vec{a}_{B,g}| = \frac{9 \times 9}{9} = 9$

$\therefore \quad x = 9$

Ans.: 9

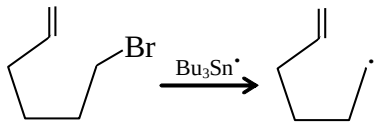
CHEMISTRY

SECTION – I
Straight Objective Type

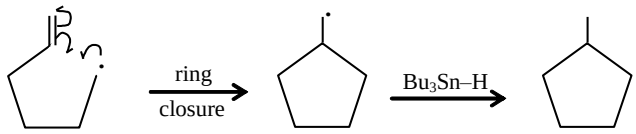
1. **(d)** Tributyl stannane reduces alkyl halide to corresponding alkane by free-radical mechanism.

CIS: $Me_2C(CN)–N=N–C(CN)Me_2 \xrightarrow{\Delta} N_2\uparrow + Me_2\dot{C}(CN)$

$Me_2\dot{C}(CN) + Bu_3Sn–H \longrightarrow Bu_3S\dot{n} + Me_2CH(CN)$

CPS: 

$+ Bu_3Sn–Br$



$+ Bu_3S\dot{n}$

Ease of formation of a five membered ring is greater than that of a six membered ring.

2. **(b)** $-\frac{d[A]}{dt} = \frac{k_1[A]}{1+k_2[A]}$

At high concentration of A ([A] >> k_2), $1 + k_2[A] \simeq k_2[A]$

$-\frac{d[A]}{dt} = \frac{k_1}{k_2} = k.$

Thus, order of the reaction is zero.

At low concentration of A([A] << k_2), $1 + k_2[A] = 1$

$-\frac{d[A]}{dt} = \frac{k_1[A]}{1} = k_1[A]$

The order of the reaction is unity.

3. **(a)** $n = \frac{N}{N_A} = \frac{1.004\times10^{23}}{6.023\times10^{23}} = \frac{1}{6}.$

$T = \frac{PV}{nR} = \frac{1\times1}{\frac{1}{6}\times0.0821} = \frac{6}{0.0821} = 73.08\ K \simeq 73\ K.$

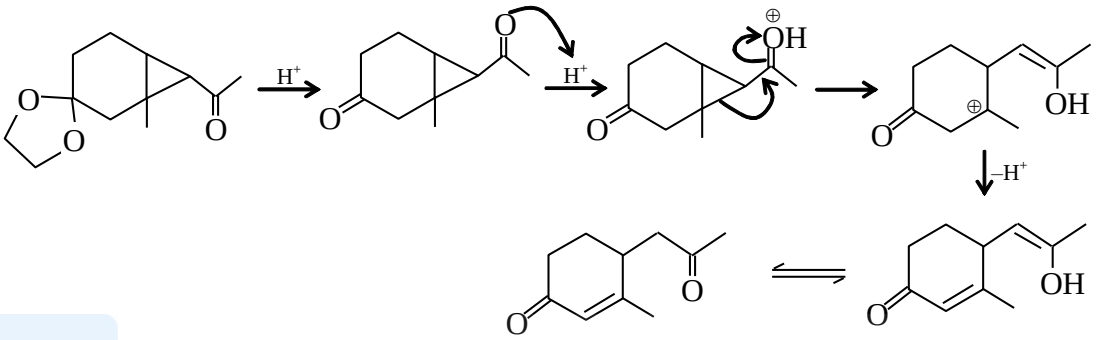
$\nu_{RMS} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3\times8.314\times73}{2\times10^{-3}}} = \sqrt{\mathbf{9.1\times10^5}}\ \mathbf{ms^{-1}}.$

4. **(c)** The second ionization energy of M refers to the reaction, $M^+(g) \longrightarrow M^{2+}(g) + e^-$

$$M(g) \longrightarrow M^{2+}(g) + 2e^- \quad \ldots. \Delta H_5$$
$$M^+(g) + e^- \longrightarrow M(g) \quad \ldots. -\Delta H_3$$
$$M^+(g) \longrightarrow M^{2+}(g) + e^- \quad \Delta H_{IE_2} = \Delta H_5 - \Delta H_3$$

SECTION – II
Multiple Correct Choice Type

5. **(a, d)** The reaction is an example of ring opening of cyclopropane in presence of acid.



6. **(a, b, c)**

$$\underset{10\text{ mL}}{HA} + \underset{10\text{ mL of } 2\times10^{-3}\text{ M}}{NaOH} \longrightarrow NaA + H_2O$$

$$[HA] = \frac{10\times2\times10^{-3}}{10} = 2\times10^{-3}\text{ M}$$

$$\underset{2\times10^{-3}(1-0.2)}{HA(aq)} + H_2O \rightleftharpoons \underset{2\times10^{-3}\times0.2}{A^-(aq)} + \underset{2\times10^{-3}\times0.2}{H_3O^+(aq)}$$

$$K_a = \frac{2\times10^{-3}(0.2)^2}{0.8} = 1\times10^{-4}$$

$$[H_3O^+]_{HA} = 2\times10^{-3}\times0.2 = 4\times10^{-4}$$

$$pH = -\log(4\times10^{-4}) = 4 - \log 4 = 3.4$$

After the neutralization of weak acid (HA) with NaOH, salt (NaA) is formed. NaA gets hydrolysed as

$$\underset{1\times10^{-3}(1-h)}{A^-(aq)} + H_2O \rightleftharpoons \underset{1\times10^{-3}\,h}{HA(aq)} + \underset{1\times10^{-3}\,h}{OH^-(aq)}$$

$$K_h = \frac{K_w}{K_a} = \frac{1\times10^{-14}}{1\times10^{-4}} = 1\times10^{-10}$$

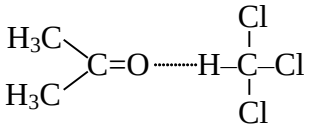
$$K_h = \frac{ch^2}{1-h} \simeq ch^2 \;;\; h = \sqrt{\frac{K_a}{c}} = \sqrt{\frac{1\times10^{-10}}{1\times10^{-3}}}$$

$$[OH^-] = 1\times10^{-3}\times\sqrt{10}\times10^{-4} = \sqrt{10}\times10^{-7}\text{ M}$$

$$pH = -\log\left(\sqrt{10}\times10^{-7}\right) = 6.5$$

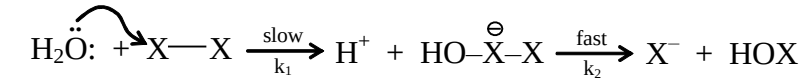
$$pH = 7.5$$

7. **(b, d)**



The mixture of acetone and chloroform show negative deviation because the interaction between acetone and $CHCl_3$ is stronger than that between acetone alone. Pure acetone cannot be obtained from the mixture at azeotropic composition.

8. **(a, c)** The mechanism of the reaction is as follows:



Rate = $k_1[X_2]\,[H_2O] = k_1'[X_2]$

where k_1' is a pseudo first order rate constant with units of sec^{-1} ($time^{-1}$). It is a displacement reaction. H–O–X is not a linear molecule but has a bent shaped structure.

SECTION – III
Reasoning Type

9. **(c)** The product of the reaction of aqueous $FeSO_4$ and NO is $[Fe(H_2O)_5NO]SO_4$, in which Fe is in +1 oxidation state and NO is also in +1 form. It has sp^3d^2 hybridisation.

Paramagnetic species should contain unpaired electron(s) but total number of electrons may be even in the species (for example O_2, S_2).

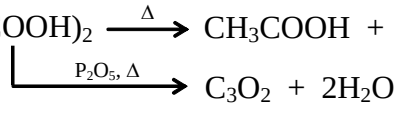
10. **(c)** For monoatomic gas ; $C_v = \frac{3}{2}RT$, $C_p = \frac{5}{2}RT$

For diatomic gas; $C_v = \frac{5}{2}RT$, $C_p = \frac{7}{2}RT$

$$C_v \text{ for mix} = \frac{1\times\frac{3}{2}RT + 1\times\frac{5}{2}RT}{2}$$

$$C_p \text{ for mix} = \frac{1\times\frac{5}{2}RT + 1\times\frac{7}{2}RT}{2} \quad ; \quad \left(\frac{C_p}{C_v}\right)_{mix} = 1.5$$

11. **(c)**



1,3-dioic acid $\xrightarrow[-CO_2]{\Delta}$ Monocarboxylic acid

1,4 and 1,5-dioic acid $\xrightarrow[-H_2O]{\Delta}$ Cyclic anhydride

1,6 and 1,7-dioic acid $\xrightarrow[-CO_2, -H_2O]{\Delta}$ Cyclic ketone

Malonic acid undergoes decarboxylation on heating alone as the reaction proceeds via 6-membered transition state while heating with P_2O_5 promotes loss of H_2O as P_2O_5 is a good dehydrating agent.

12. (a)

SECTION–IV
Comprehension Type

Passage-I

13. (b) $s = [Al^{3+}] + [Al(OH)_4^-]$

$$s = \frac{K_{sp}}{[OH^-]^3} + K_c [OH^-]$$

For minimum solubility, differentiate s w.r.t. $[OH^-]$ and set it equal to zero.

$$\frac{ds}{d[OH^-]} = \frac{-3K_{sp}}{[OH^-]^4} + K_c = 0$$

$$\frac{3K_{sp}}{[OH^-]^4} = K_c$$

$$[OH^-] = \left(\frac{3K_{sp}}{K_c}\right)^{\frac{1}{4}}$$

14. (b) Solubility of $Fe(OH)_3$ decreases with increase in concentration of $[OH^-]$ or decrease in concentration of $[H^+]$.

Addition of further amount of NH_4Cl will increase the concentration of NH_4^+ and thus the ionization of NH_4OH is suppressed. This lowers the $[OH^-]$ concentration and thus pOH increases or pH decreases.

In NH_4I solution, NH_4^+ ion hydrolyses giving H_3O^+, thus making the solution acidic while in HCO_2K solution, HCO_2^- ion hydrolyses to give OH^-, thus solution is basic in nature.

15. (b) $AgCl(s) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq) + Cl^-(aq)$

$$K_1 = K_f \times K_{sp}(AgCl) = 1.6 \times 10^{-10} \times 10^7 = 1.6 \times 10^{-3} = \frac{[(Ag(NH_3)_2^+][Cl^-]}{[NH_3]^2}$$

$AgCN(s) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq) + CN^-(aq)$

$K_2 = K_f \times K_{sp}(AgCN) = 2.5 \times 10^{-16} \times 10^7 = 2.5 \times 10^{-9} = \dfrac{[(Ag(NH_3)_2^+][CN^-]}{[NH_3]^2}$

$\dfrac{[Cl^-]}{[CN^-]} = \dfrac{1.6\times10^{-3}}{2.5\times10^{-9}} = 6.4 \times 10^5$; $K_1 = \dfrac{x^2}{(1-2x)^2}$; $x = [Cl^-] =$ **0.037 M**

$[CN^-] = \dfrac{0.037}{6.4\times10^5} =$ **5.78×10^{-8} M.**

Passage-II

16. **(b)** The optical purity of racemized alcohol $= \dfrac{[\alpha]_{mixture}}{[\alpha]_{pure}} \times 100$

$= \dfrac{-6}{-10} \times 100 =$ **60%**.

17. **(c)** Percentage inversion $= \dfrac{\text{Optical purity of alcohol}}{\text{Optical purity of bromide}} \times 100$

Now, optical purity of bromide $= \dfrac{[\alpha]_{mixture}}{[\alpha]_{pure}} \times 100$

$= \dfrac{30}{36} \times 100 = 83\%.$

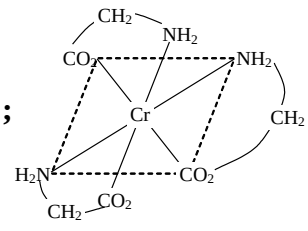
$\therefore$ Percentage inversion $= \dfrac{60}{83} \times 100 =$ **72%**.

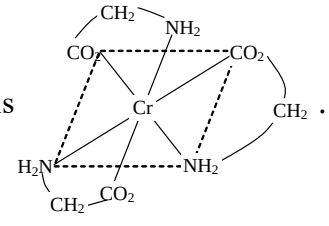
18. **(a)** Total inversion = Pure inverted product + Half of racemized product

$= 72 + \left(\dfrac{1}{2} \times 28\right) = 86\%$

$\therefore$ Percentage of front side attack $= \dfrac{1}{2} \times 28 =$ **14%**.

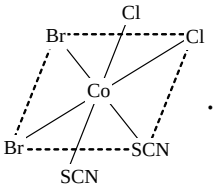
SECTION–IV
Matrix–Match Type

1. **(A) – 2, 3, 4, 5 ;** 

is optically active due to absence of a plane of symmetry and centre of symmetry. It's geometrical isomer is 

.

Cr^{3+} has d^3 configuration, so magnetic moment will be $\sqrt{15}$ and is octahedral in shape.

(B) – 1, 2, 3, 4, 5 ; It can show linkage isomerism due to SCN ligand. Co^{2+} has $3d^7$ configuration, so μ_s will be $\sqrt{15}$. It can show geometrical isomerism and has octahedral geometry. Its optically active isomer will be 

.

(C) – 5 ; Its an octahedral complex.

(D) – 1, 4 ; Linkage isomerism is shown due to NO_2 ligand. It is an square planar complex, which is always optically inactive. It can show geometrical isomerism (3 isomers).

2. **(A) – 1, 4, 5 ;** (A) has a chiral centre and can show geometrical isomerism. Stereogenic units are structural units around which stereoisomers can be generated.

(B) – 2, 3, 5 ; (B) is optically inactive due to presence of plane of symmetry and it has 2 stereogenic units.

(C) – 1, 4 ; (C) is optically active as it has one chiral centre.

(D) – 2, 3, 5 ; (D) is optically inactive due to symmetry (plane of symmetry) and possesses two stereogenic units.

mathongo

MATHEMATICS

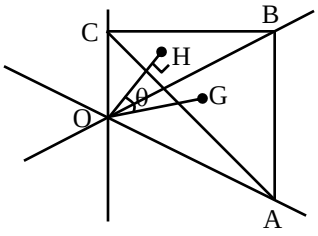
SECTION – I

Single Correct Choice Type

1. **(b)** $\Delta^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2$

$$= \left(\frac{1}{2}\frac{p^2}{lm}\right)^2 + \left(\frac{1}{2}\frac{p^2}{mn}\right)^2 + \left(\frac{1}{2}\frac{p^2}{ln}\right)^2$$

$$\Rightarrow \Delta = \frac{p^2}{2lmn}$$



Also $\left(\frac{l^2+m^2+n^2}{3}\right)^3 \geq l^2m^2n^2 \quad \because A.M. \geq G.M.$

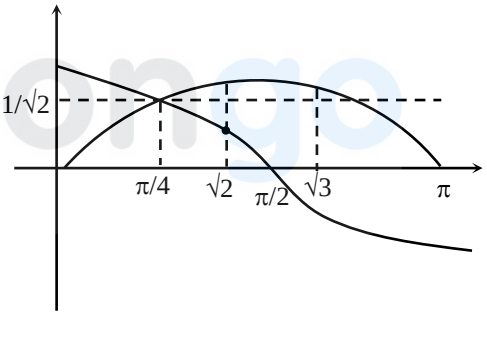
$$\Rightarrow \frac{p^4}{4(lmn)^2} \geq \frac{27p^4}{4} \Rightarrow \Delta \geq \frac{3\sqrt{3}p^2}{2}$$

2. **(c)** The number of common vertices are the number of common roots to the equation $Z^{1982} - 1 = 0$ and $Z^{2973} - 1 = 0$, which is GCD of 1982, 2973, i.e., 991

3. **(a)** As $\sqrt{2} < \frac{\pi}{2} < \sqrt{3} \Rightarrow \sin\sqrt{2} > 0$ and

$\cos\sqrt{3} < 0$

$\Rightarrow \sin\sqrt{2} - \cos\sqrt{3} > 0$



Also $\sqrt{3} < \frac{3\pi}{4} \Rightarrow \sin\sqrt{3} > \frac{1}{\sqrt{2}}$

$\sqrt{2} > \frac{\pi}{4} \Rightarrow \cos\sqrt{2} < \frac{1}{\sqrt{2}}$

$\Rightarrow -\cos\sqrt{2} + \sin\sqrt{3} > 0$

$\therefore$ Curve is an ellipse of form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$= \left|\sin\sqrt{2} - \cos\sqrt{3}\right| > \left|\sin\sqrt{3} - \cos\sqrt{2}\right|$

4. **(d)** $\frac{r}{4r^4+1} = \frac{r}{(2r^2-2r+1)(2r^2+2r+1)}$

$$= \frac{1}{4}\left[\frac{(2r^2+2r+1)-(2r^2-2r+1)}{(2r^2-2r+1)(2r^2+2r+1)}\right]$$

$$= \frac{1}{4}\left[\frac{1}{(2r^2-2r+1)} - \frac{1}{(2r^2+2r+1)}\right]$$

$$\Rightarrow \sum_{r=1}^{n} \frac{r}{4r^4+1} = \frac{1}{4}\left[1 - \frac{1}{2n^2+2n+1}\right] = \frac{\sum n}{4\sum n + 1}$$

5. **(c)** $f(x)=e^{x-k}$ then inverse of $f(x)$; $f^{-1}(x)=(\ln x)+k$,
and also both functions are increasing, therefore
$f(x)=f^{-1}(x)$ is equivalent to $f(x)=f^{-1}(x)=x$
$\Rightarrow \quad \ln x+k=x$ should have a solution
$\Rightarrow \quad k=x-\ln x$
Now, let $g(x)=x-\ln x$
has least value 1 as $g'(x)=1-\dfrac{1}{x}$ has a minimum at $x=1$
and $\lim\limits_{x\to 0^+} g(x)$, $\lim\limits_{x\to\infty} g(x)$ both approach to ∞.

6. **(a)** $(A-B)C=BA^{-1}$
$\Rightarrow \quad AC-BC-BA^{-1}+AA^{-1}=I_3$
$\Rightarrow (A-B)(C+A^{-1})=I_3 \quad \Rightarrow \quad (A-B)^{-1}=C+A^{-1}$
$\Rightarrow (C+A^{-1})(A-B)=I_3 \quad \Rightarrow \quad C(A-B)+A^{-1}A-A^{-1}B=I_3$
$\Rightarrow C(A-B)=A^{-1}B$

7. **(c)** Given every element of set *B* has been mapped to some element of set *A*.
Without loss of generality we can assume
$b_1<b_2<b_3.....<b_n$
Consider a_1, a_2, ...a_{2n} be divided into *n* sub-groups, so that each b_i is mapped to one sub group and all elements mapped to a particular b_i to hold equality.
Therefore the correct answer is the number of ways in which $2n$ elements can be divided into *n* non-empty sub groups i.e., ${}^{2n-1}C_n$.

8. **(c)** Let $x_n=\tan\theta_n$

Now, $\tan\theta_{n+1}=x_{n+1}=\dfrac{x_n}{1+\sqrt{1+x_n^2}}=\dfrac{\tan\theta_n}{1+\sqrt{1+\tan^2\theta_n}}$

$\Rightarrow \quad \tan\theta_{n+1}=\dfrac{\tan\theta_n}{1+\sec\theta_n}=\dfrac{\sin\theta_n}{1+\cos\theta_n}=\tan\dfrac{\theta_n}{2}$

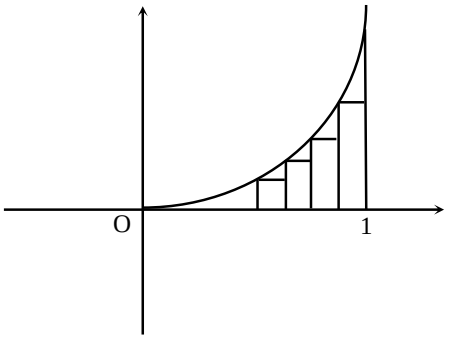
$\Rightarrow \quad \theta_{n+1}=\dfrac{\theta_n}{2} \Rightarrow \theta_n=\dfrac{\theta_{n-1}}{2}$

Now, $\theta_1=\dfrac{\pi}{3} \Rightarrow \theta_n=\dfrac{\pi}{3.2^{n-1}} \Rightarrow x_n=\tan\left(\dfrac{2\pi}{3.2^n}\right)$

$\Rightarrow \quad \lim\limits_{n\to\infty} 2^n x_n=\lim\limits_{n\to\infty}\dfrac{\tan(2\pi/3.2^n)}{(1/2^n)}=\dfrac{2\pi}{3}$

SECTION – II
Multiple Correct Choice Type

9. (a), (c) & (d)

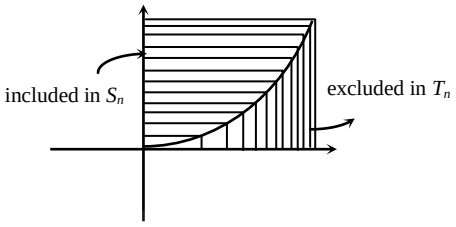


$$S_n = \lim_{n\to\infty}\left(\frac{1}{n^6}+\frac{32}{n^6}+\frac{243}{n^6}+.....\frac{1}{n}\right)$$

$$T_n = \lim_{n\to\infty}\left(\frac{1}{n^6}+\frac{32}{n^6}+\frac{243}{n^6}+.....\frac{(n-1)^5}{n^6}\right)$$

$$S_n = \lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^5}{n^6} = \int_0^1 x^5\ dx = \frac{x^6}{6}\Big|_0^1 = \frac{1}{6}$$

$$\Rightarrow\quad S_n = \left(\frac{1}{6}\right)^+;\ T_n = \left(\frac{1}{6}\right)^-$$



But since x^5 is concave upward the area included in S_n for any two consecutive values of r is more than area excluded in T_n for same values of r.

$$\Rightarrow\quad S_n - \frac{1}{6} > \frac{1}{6} - T_n \quad\Rightarrow\quad S_n + T_n > \frac{1}{3}$$

10. (b), (c)

Let $S = \sum_{j=1}^{n}\sum_{k=1}^{n}\frac{z_j}{z_k} = (z_1 + z_2 + ...z_n)\left(\frac{1}{z_1}+\frac{1}{z_2}+.....\frac{1}{z_n}\right)$

If $|z_1| = |z_2| .....|z_n| = r$

$$\Rightarrow\quad \frac{1}{z_k} = \frac{\bar{z}_k}{r^2}$$

$$\Rightarrow\quad S = \frac{1}{r^2}(z_1 + z_2 + ...z_n)(\bar{z}_1 + \bar{z}_2 + ...\bar{z}_n)$$

$$= \frac{|z_1 + z_2 + ...z_n|^2}{r^2}$$

But $z_1 + z_2 + ...z_n = |z| = r$

$$\Rightarrow\quad S = \frac{r^2}{r^2} = 1$$

11. **(b)** $(f'(x))^2 - k^2(f(x))^2 \le 0$

$\Rightarrow \quad (f'(x) - kf(x))\,(f'(x) + kf(x)) \le 0$

$\Rightarrow \quad \left(f(x)e^{-kx}\right)' \left(f(x)e^{kx}\right)' \le 0$

$\Rightarrow$ Exactly one of the functions $g_1(x) = f(x)e^{-kx}$ or $g_2(x) = f(x)e^{kx}$ is non decreasing.

But $f(0) = 0 \Rightarrow$ both functions g_1 and g_2 have a value zero at $x = 0$

$\forall x \in [0, 1]$, $g_1(0) = 0$ and g_1 increasing $\Rightarrow \; g_1(x) \ge 0 \;\Rightarrow\; f(x) \ge 0$

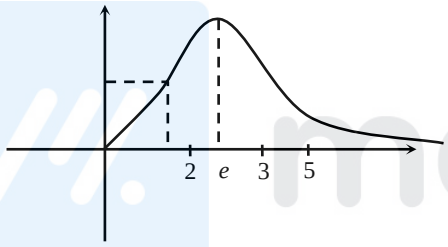
$g_2(0) = 0$ and g_2 decreasing $\Rightarrow \; g_2(x) \le 0 \;\Rightarrow\; f(x) \le 0$

$\Rightarrow \quad f(x) = 0 \; \forall x \in [0, 1]$

12. **(b), (c) & (d)**

Solving we get $x = 2^{1/2}$, $y = 3^{1/3}$, $z = 5^{1/5}$

Using graph of $x^{1/x}$



$\Rightarrow \quad 3^{1/3} > 5^{1/5}$

Also $2^{1/2} < 3^{1/3}$	$2^{1/2} > 5^{1/5}$
as $2^3 < 3^2$	$2^5 > 5^2$

$\Rightarrow \quad y > x > z$

Hence (b), (c) & (d) are NOT true.

SECTION–III
Comprehension Type

13. **(c)** Normal at point P is angle bisector of angle SPS' (which is also obvious as angle bisector of $\angle BPC$ is common tangent to E_1 and E_2).

Since angle bisector of AC and BD are perpendicular E_3 is orthogonal to E_1 and E_2 at P.

For E_3, $2ae = BC$ and $2a = BP + PC$

$\Rightarrow \quad e = \dfrac{BC}{BP + PC}$

Moreover if circumcircle of ΔBPC would have touched ellipse, angle bisector of BPC would have been normal to circle which will be true if BPC is isosceles.

14. **(b)** $e_1 = \dfrac{AB}{PA+PB}$; $e_2 = \dfrac{CD}{PC+PD}$ and $e_3 = \dfrac{BC}{PB+PC}$

Since ΔPAB and ΔPDC are similar $\dfrac{PB}{PC} = \dfrac{PA}{PD} = \dfrac{AB}{CD} = \lambda$

$\Rightarrow \; e_1 = e_2$

15. **(a)** $MB = \dfrac{AB(PB)}{PB+PA} = PBe_1$

$CN = PCe_2 = PCe_1$

Adding $MB + CN = e_1(PB+PC) = \dfrac{e_1 BC}{e_3} \quad \Rightarrow \quad \dfrac{MB+CN}{BC} = \dfrac{e_1}{e_3}$

SECTION–IV
Matrix–Match Type

1. **A-2** $f''(0) = \dfrac{1+0}{1+f(0)} = \dfrac{1}{2} \;\Rightarrow\; f'(x)$ is increasing at $x = 0$

$\Rightarrow$ for some $\delta > 0$ $f'(x) > 0$, $\forall\, x \in (0, \delta)$

$\Rightarrow$ $f(\delta) > f(0) \;\Rightarrow\; f(\delta) > 1$ …(i)

$f''(\delta) = \dfrac{1+\delta}{1+f(\delta)} > 0 \;\Rightarrow\; f'(x)$ is increasing at $x = \delta \Rightarrow\; f'(\delta) > 0$

Hence $f(x)$ is increasing at $x = \delta$

Repeat the same argument on interval right to $x = \delta$.

It comes out $f'(x) > 0$, $\forall x > 0$.

Hence $f(x)$ is increasing $\forall x > 0$.

B-2 $f(x) = e^x$ which is increasing function.

C-2, 4 If $f(x)$ is polynomial function of degree n, then

$h(x) = f(x) - x$ is also a polynomial of degree n

Now, $f(x^2+1) = f^2(x) + 1$

$x = 0 \;\Rightarrow\; f(1) - 1 = 0 \;\Rightarrow\; h(1) = 0$

$x = 1 \;\Rightarrow\; f(2) - 2 = 0 \;\Rightarrow\; h(2) = 0$

……..

……

$x = r \;\Rightarrow\; f(r) - r = 0 \;\Rightarrow\; h(r) = 0$ for $r \in N$

Keep on repeating it until $r > n$

$\Rightarrow$ $h(x) = 0$ has more than n roots

Hence $h(x) = 0$, $\forall\; x \in R$ (it's an identity)

$\Rightarrow$ $f(x) = x$, $\forall\, x \in R$ i.e., linear and increasing.

D-2, 4 From given functional equation it is obvious that $f(x)$ has to be a linear polynomial.

Let $f(x)=ax+b$

$\Rightarrow \quad f(ax+b)=6x-(ax+b) \quad \Rightarrow \quad a(ax+b)+b=6x-ax-b$

$\Rightarrow \quad x(a^2-b+a)=0$ and $ab+2b=0$

Solving these equation, $a=2$ and $b=0$ or $a=-3$ and $b=0$

$\Rightarrow \quad f(x)=2x$ or $f(x)=-3x$

But $f(x)=-3x$ is not $R^+ \to R^+$

Hence $f(x)=2x$ i.e., linear and increasing.

2. **(A)–2; (B)–1; (C)–1; (D)–3**

Total number of triangles equals to ${}^8C_3=56$

Total number of equilateral triangles whose one vertex is A equals to 3. ($\Delta AFH, \Delta AFC$ and ΔAHC). Now total number of vertices equals to 8 and all the triangle made by three vertices, so total number of equilateral triangles are equal to $\frac{8\times 3}{3}=8$

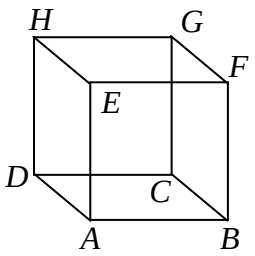
Total number of isosceles triangle in plane $ABCD$ is 4.

$\Rightarrow$ Total number of isosceles triangles (not equilateral) are equal to 4 × 6 = 24

$\Rightarrow$ Now total number of scalene triangles $={}^8C_3-24-8=24$

Now, all proper isosceles triangle (not equilateral) are right angled triangles. Because scalene triangle is formed by one edge, one force diagonal and one cube diagonal, so all scalene triangle are right angle triangled.

$\Rightarrow$ Total right angled triangle = 24 + 24 = 48



SECTION–V
Integer Answer Type

1. Given $g(x_1)g(x_2)g(x_3)g(x_4)g(x_5)=A$

$=(x_1^2-2)(x_2^2-2)(x_3^2-2)(x_4^2-2)(x_5^2-2)$

$=-(2-x_1^2)(2-x_2^2)(2-x_3^2)(2-x_4^2)(2-x_5^2)$ ..(i)

$=-[2^5-(\sum x_1^2)2^4+\sum x_1^2.x_2^2.2^3-\sum x_1^2.x_2^2.x_3^2.2^2$

$+\sum x_1^2.x_2^2.x_3^2.x_4^2.2-x_1^2.x_2^2\ x_3^2.x_4^2.x_5^2\ ]$

$p(x)=x^5+x^2+1=0$ has roots $x_1,\ x_2,\ ....,x_5$, then that equation $q(x)$ whose roots are square of the roots of $p(x)$ is $q(x)=(\sqrt{y})^5+(\sqrt{y})^2+1=0$; $\alpha=x$ and $y=\alpha^2$

$\Rightarrow \quad (y+1)^2 = \left(-\sqrt{y}\right)^{5\times 2}$

$\Rightarrow \quad y^2+2y+1=y^5 \quad \Rightarrow \quad q(x)=y^5-y^2-2y-1=0,$

then $\sum x_1^2 = \sum y_1 = 0$

$\sum x_1^2.x_2^2 = \sum y_1.y_2 = 0$

$\sum x_1^2.x_2^2.x_3^2 = \sum y_1.y_2.y_3 = 1$

$\sum x_1^2.x_2^2.x_3^2.x_4^2 = \sum y_1.y_2.y_3.y_4 = -2$

$\sum x_1^2.x_2^2.x_3^2.x_4^2.x_5^2 = \sum y_1.y_2.y_3.y_4.y_5 = 1$, then

$A = -[2^5 - 0 + 0 - 2^2 - 2.2 - 1] = -[32-4-4-1] = -[32-9]$

$= -23$

$x_1x_2x_3x_4x_5 = -1 \Rightarrow g(x_1x_2.....x_5) = -1$

$\Rightarrow g(x_1)g(x_2).....g(x_5) - 30g(x_1x_2.....x_5) = 7$

Ans. 7

Alternative:

Let us form that equation having roots $y = g(x_i)$ i.e., $y = x^2 - 2$

$x = \sqrt{y+2}$

$\Rightarrow \quad \left(\sqrt{y+2}\right)^5 + \left(\sqrt{y+2}\right)^2 + 1 = 0$

$\Rightarrow \quad y^5 + 20y^4 + 40y^3 + 79y^2 + 74y + 23 = 0$

$\therefore \quad g(x_1).....g(x_5)$ = Product of roots

$= -23$

$x_1x_2x_3x_4x_5 = -1 \Rightarrow g(x_1x_2.....x_5) = -1$

$\Rightarrow g(x_1)g(x_2).....g(x_5) - 30g(x_1x_2.....x_5) = 7$

Ans. 7

2 . $f(x)+f(\alpha x)+f(\alpha^2 x)+.....+f(\alpha^6 x) = 7A_0 + \sum_{k=1}^{20} A_k x^k \left(1+\alpha^k+.....+\alpha^{6k}\right)$

but when $k \neq 7$ and $k \neq 14$, then $1+\alpha^k+\alpha^{2k}+....+\alpha^{6k} = 0$

Hence $f(x)+f(\alpha x)+.....+f(\alpha^6 x) = 7A_0 + 7A_7x^7 + 7A_{14}x^{14}$

$= 7\left(A_0 + A_7x^7 + A_{14}x^{14}\right)$

$k = 7$

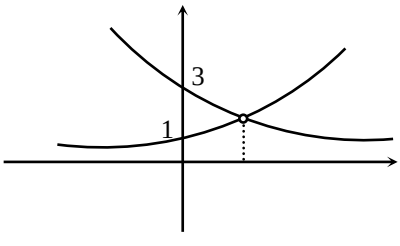
Ans. 7

3. Let $f^{-1}(x)$ be $g \Rightarrow \ln\left(g+\sqrt{g^2+1}\right) = x$

$\Rightarrow \quad g+\sqrt{g^2+1} = e^x \qquad \ldots(i)$

and $-g+\sqrt{g^2+1} = e^{-x} \qquad \ldots.(ii)$

Now, $\left|e^x - e^{-x}\right| = 2e^{-|x|}$



Case I: $x > 0$; $e^{-|x|} = e^{-x}$ and $e^{x} > e^{-x}$

$\Rightarrow \; e^{x} - e^{-x} = 2e^{-x} \;\Rightarrow\; e^{x} = 3e^{-x}$

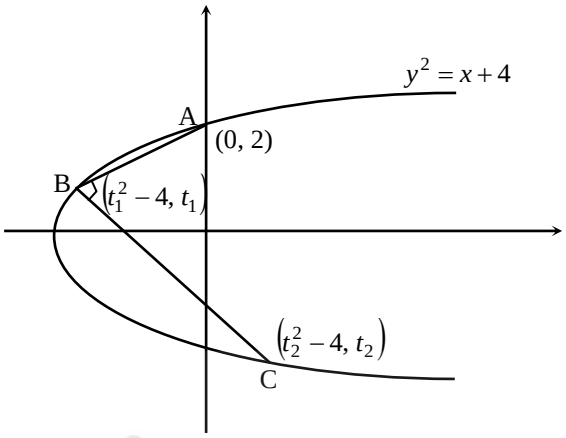
Case II: $x < 0$; $e^{-|x|} = e^{x}$ and $e^{x} < e^{-x}$

$\Rightarrow \; e^{-x} - e^{x} = 2e^{x} \;\Rightarrow\; e^{-x} = 3e^{x}$

$\therefore$ Two solutions.

Ans. 2

4. Now $\left(\frac{t_2 - t_1}{t_2^2 - t_1^2}\right)\left(\frac{t_1 - 2}{t_1^2 - 4}\right) = -1$



put $t_1 \neq t_2$ and $t_1 \neq 2$

$\Rightarrow \; \frac{1}{(t_1 + t_2)}\frac{1}{(t_2 + 2)} = -1$

$\Rightarrow \; t_1^2 + t_1 t_2 + 2t_1 + 2t_2 + 1 = 0$

Quadratic in t_1 should have real roots.

$\Rightarrow \; (t_2 + 2)^2 - 4(2t_2 + 1) \geq 0 \;\Rightarrow\; t_2^2 - 4t_2 \geq 0$

$\Rightarrow \; t_2 \leftarrow (-\infty, 0] \cup [4, \infty)$

There least positive value of t_2 is 4.

Ans. 4